Analytical Problem Solving

CO-2, AT-1

COMPUTER VISION

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Thresholding and Intensity-Level Slicing

Given,

Input intensity values:-

$$r = [25, 60, 95, 130, 165, 200, 235]$$

Threshold value : $T = 128$

1. Binary Thresholding

Formula:-

$$S = \begin{cases} 255 & r \geq 128 \\ 0 & r < 128 \end{cases}$$

Calculation:-

- $25 < 128 \rightarrow 0$
- $60 < 128 \rightarrow 0$
- $95 < 1280 \rightarrow 0$
- $130 \geq 128 \rightarrow 255$
- $165 \geq 128 \rightarrow 255$
- $200 \geq 128 \rightarrow 255$
- $235 \geq 128 \rightarrow 255$

Output :-

$[0, 0, 0, 255, 255, 255, 255]$

2. Intensity-Level Slicing

Intensity range is 90 to 170. Values within this range are changed to 255, while the remaining values stay unchanged.

Output :- $[25, 60, 255, 255, 255, 200, 25]$

3. percentage of Highlighted pixels

No. of highlighted pixels = 3

Total pixels = 7

$$\text{percentage} = \frac{3}{7} \times 100 = 42.86\%$$

4. Mean Intensity

Original mean: $\frac{25+60+95+130+165+200+235}{7} = 126$

Mean after intensity slicing: $\frac{25+60+255+255+255+200+235}{7} = 183.57$

Binary thresholding Seperates pixels into two intensity levels. Intensity slicing highlights the selected range and make the required intensity values easier to identify.

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2) piecewise contrast stretching

Given Control points: $(r_1, s_1) = (70, 20)$

$(r_2, s_2) = (140, 230)$

Maximum intensity:

$L = 256$

Input values:

$r = [30, 70, 90, 110, 140, 200]$

Assume additional endpoints are (0,0) and (255, 255)

1. Calculate the slopes

First Segment : 0 to 70

$$m_1 = \frac{20-0}{70-0} = \frac{2}{7}$$

$$\therefore S = \frac{2r}{7}$$

Second segment : 70 to 140

$$m_2 = \frac{230-20}{140-70} = \frac{210}{70} = 3$$

$$\therefore S = 20 + 3(r-70)$$

$$= 3r - 190$$

Third Segment : 140-255

$$m_3 = \frac{255-230}{255-140} = \frac{25}{115} = \frac{5}{23}$$

$$\therefore S = 230 + \frac{5}{23}(r-140)$$

2. calculate Transformed values

Input r	Calculation	Output S
30	2(30)/7	9
70	Given control point	20
90	3(90) - 190	80
110	3(110) - 190	140
140	Given control point	230
200	230 + (5/23)(60)	243

Final Output : [9, 20, 80, 140, 230, 243]

3. Contrast Analysis

Middle region input width: 140 - 70 = 70

Middle region output width: 230 - 20 = 210

Contrast gain: $\frac{210}{70} = 3$

3) Spatial Filtering

1. Mean

Given Image $\begin{bmatrix} 52 & 55 & 58 \\ 54 & 255 & 57 \\ 53 & 56 & 54 \end{bmatrix}$

The centre pixel 255 is considered a noisy pixel

1. Mean Filter

Formula: $\text{Mean} = \frac{\text{Sum of all pixels}}{\text{No. of pixels}}$

$= \frac{52+55+58+54+255+57+53+56+54}{9}$

$= \frac{699}{9} = 77.67$

$\therefore$ **Mean = 77.67**

2. Weighted Average Filter

Given Mask:

$$\frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

Weighted Sum = 1686

Weighted Avg = $\frac{1686}{16}$

$= \boxed{105.375}$

3. Median Filter

Arrange pixel values in ascending order:

52, 53, 54, 55, 56, 57, 58, 59, 254

Median = 56 (Middle value)

4. Error Calculation

Avg of eight surrounding pixels: $\frac{52+55+58+54+57+53+56+59}{8}$

$= \frac{444}{8} = \boxed{55.5}$

Absolute error for mean filter

$|177.67 - 55.5| = 22.17$

Absolute error for weighted avg filter:

$|105.375 - 55.5| = 49.875$

Absolute error for median filter:

$|56 - 55.5| = 0.5$

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4) Edge Detection using Sobel and prewitt operators

Given Image $I = \begin{bmatrix} 10 & 10 & 10 \\ 10 & 10 & 90 \\ 10 & 90 & 90 \end{bmatrix}$

1. Sobel operator

Given $G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$

$G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$

Calculating horizontal gradient: $G_x = 240$

Calculating vertical gradient: $G_y = 240$

- Gradient magnitude: $M = \sqrt{G_x^2 + G_y^2}$

$$M = \sqrt{240^2 + 240^2}$$

$$\therefore \boxed{M = 339.41}$$

Approximate magnitude:

$$|G_x| + |G_y| = 240 + 240 = 480$$

percentage error: $\frac{480 - 339.41}{339.41} \times 100 = 41.42\%$

Gradient director: $\theta = \tan^{-1}\left(\frac{G_y}{G_x}\right)$

$$\boxed{\theta = 45^\circ}$$

2. prewitt operation

Given

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}$$

$$G_y = \begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

Calculating gradients:

$$G_x = 160, \quad G_y = 160$$

Gradient magnitude: $M = \sqrt{160^2 + 160^2}$

$$\boxed{M = 226.27}$$

Approximate magnitude:

$$|G_x| + |G_y| = 160 + 160 = 320$$

percentage error: $\frac{320 - 226.27}{226.27} \times 100 = \boxed{41.42\%}$

Gradient direction: $\boxed{\theta = 45^\circ}$

3) Comparison

Given edge detection threshold:

$$T = 250$$

- Sobel magnitude = 339.41, which is greater than 250. Therefore, an edge is detected.
- prewitt magnitute = 226.27, which is lose than 250. Therefore, an edge is not detected.

5) Image Thresholding Using Histogram Analysis

Given,

Total no of pixels = 32

									Total
Intensity	0	1	2	3	4	5	6	7	
Frequency	3	6	4	1	1	5	8	4	32

1. Calculate Global Mean

Formula:-

$$\mu = \frac{\sum (\text{Intensity} \times \text{Frequency})}{N}$$

$$\mu = \frac{0(3) + 1(6) + 2(4) + 3(1) + 4(1) + 5(5) + 6(8) + 7(4)}{32}$$

$$\mu = \frac{122}{32} \Rightarrow \boxed{\mu = 3.8125}$$

2. Iterative Thresholding

Initial threshold:-

$$T_0 = 3.8125$$

class 1 contains intensity levels 0 to 3

$$m_1 = \frac{0(3)+1(6)+2(4)+3(1)}{3+6+4+1}$$

$$m_1 = \frac{17}{14} = 1.2143$$

class 2 contains intensity levels 4 to 7

$$m_2 = \frac{4(1)+5(5)+6(8)+7(4)}{1+5+8+4}$$

$$m_2 = \frac{105}{18} = 5.8333$$

New threshold:

$$T_1 = \frac{m_1+m_2}{2}$$

$$T_1 = \frac{1.2143+5.8333}{2}$$

$$\boxed{T_1 = 3.5238}$$

change in Threshold

$$|T_1 - T_0| = |3.5238 - 3.8125|$$

$$= \boxed{0.2887}$$

Since $0.2887 < 0.5$, iteration stops.

∴ Final iterative threshold:

$$\boxed{T = 3.5238}$$

3. Otsu Threshold

The calculated between-class variances are:

Threshold k	2	3	4
Btwn-class variance.	51202	52506	51354

The maximum variance occurs at

$$\boxed{k^* = 3}$$

∴ The optimum Otsu threshold is 3.

4. Binary Mapping

The image contains dark objects on a bright background.

- Intensity levels 0-3 → object → 1
- Intensity levels 4-7 → Background → 0

Intensity levels	Binary value
0, 1, 2, 3	1
4, 5, 6, 7	0

No of object pixels:

$$3 + 6 + 4 + 1 = 14$$

percentage of object pixels: $\frac{14}{32} \times 100 = \boxed{43.75\%}$

5. Comparison of Thresholding Methods

- Global mean thresholding: Simple and easy to calculate.
- Iterative thresholding: Repeatedly calculates the class means to obtain a suitable threshold.
- Otsu thresholding: Selects the threshold with maximum between-class variance.